



A

Microproject

on

Scientific Calculator

Submitted to

MSBTE

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Computer Engineering

Under I Scheme

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YASHWANTRAO BHONSALE INSTITUTE OF
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**MAHARASHTRA STATE
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On Sixth semester of diploma in **COMPUTER ENGINEERING** of institute **Yashwantrao Bhonsale Institute Of Technology (1742)** has completed the **Micro-Project** satisfactorily **on Mobile Application Development (22617)** for the academic year 2023-2024 as per prescribed in the curriculum.

Project Guide

Head of the Department

Principal

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1. Abstract

A scientific calculator is a sophisticated electronic device designed to perform various mathematical functions beyond simple arithmetic operations. It typically includes features such as trigonometric functions, logarithms, exponential calculations, and advanced operations like calculus and statistical analysis. Scientific calculators are invaluable tools for students, engineers, scientists, and professionals working in fields requiring complex mathematical computations. They provide accuracy, efficiency, and convenience in solving equations, analyzing data, and conducting experiments. With their compact design and user-friendly interface, scientific calculators have become essential instruments in both educational and professional settings, aiding in problem-solving and facilitating mathematical exploration. A scientific calculator operates on principles of mathematical algorithms and utilizes integrated circuits to execute computations with precision and speed. Its functionality extends beyond basic arithmetic to include scientific notations, fractions, and customizable settings for various modes of calculation. Advanced features such as matrices, vectors, and complex numbers enable users to solve complex mathematical problems efficiently. Additionally, scientific calculators often incorporate memory functions, allowing users to store and recall values for repetitive calculations or multi-step problem-solving. With their portability and versatility, scientific calculators have become indispensable tools in fields such as engineering, physics, chemistry, and mathematics, empowering users to perform intricate calculations with ease and accuracy.

2.Introduction

A scientific calculator is a sophisticated handheld device or software application specifically designed to handle complex mathematical calculations beyond the capabilities of basic calculators. These calculators are equipped with advanced functions such as trigonometric, logarithmic, and exponential operations, along with features for statistical analysis, algebraic manipulation, and calculus. Unlike simple calculators, scientific calculators offer users the ability to perform intricate computations commonly encountered in fields like engineering, physics, chemistry, and mathematics. With their comprehensive functionality and user-friendly interfaces, scientific calculators serve as indispensable tools for students, professionals, and researchers alike, facilitating accurate and efficient problem-solving in diverse academic and professional settings. Scientific calculators have evolved to incorporate a wide range of features, including customizable settings, memory functions, and the ability to handle complex equations with ease. They are equipped with specialized keys and symbols to navigate through various mathematical functions swiftly. Additionally, modern scientific calculators often come with user-friendly interfaces, graphical displays, and programmability options, allowing users to streamline their workflow and tailor the calculator's functionality to their specific needs. Whether used for classroom assignments, scientific research, engineering projects, or everyday calculations, scientific calculators have become indispensable companions for individuals seeking precision and efficiency in their mathematical endeavors.

3.Objective

The objective of a scientific calculator is to provide users with a powerful tool capable of performing a wide range of mathematical functions accurately and efficiently. These calculators aim to simplify complex calculations commonly encountered in fields such as mathematics, science, engineering, and finance. By offering functionalities beyond basic arithmetic operations, such as trigonometric functions, logarithms, exponentials, and statistical analysis, scientific calculators enable users to solve equations, analyze data, and perform computations with precision. Furthermore, scientific calculators often come with additional features like customizable settings, memory functions, and programmability options, allowing users to tailor the calculator to their specific needs and streamline their workflow. Ultimately, the objective of a scientific calculator is to empower individuals with a versatile tool that enhances their problem-solving capabilities and facilitates mathematical exploration in various educational, professional, and research settings.

- Provide users with a versatile tool capable of performing complex mathematical calculations accurately and efficiently.
- Simplify computations commonly encountered in fields such as mathematics, science, engineering, and finance.
- Offer functionalities beyond basic arithmetic operations, including trigonometric functions, logarithms, exponentials, and statistical analysis.
- Enable users to solve equations, analyze data, and perform computations with precision.
- Incorporate additional features such as customizable settings, memory functions, and programmability options to enhance user experience and streamline workflow.
- Empower individuals with a reliable instrument that enhances problem-solving capabilities and facilitates mathematical exploration in various educational, professional, and research settings.

4.code

Main java code

```
package com.example.myapplication

import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    // creating variables for our text view and button
    lateinit var tvsec: TextView
    lateinit var tvMain: TextView
    lateinit var bac: Button
    lateinit var bc: Button
    lateinit var bbrac1: Button
    lateinit var bbrac2: Button
    lateinit var bsin: Button
    lateinit var bcos: Button
    lateinit var btan: Button
    lateinit var blog: Button
    lateinit var bln: Button
    lateinit var bfact: Button
    lateinit var bsquare: Button
    lateinit var bsqrt: Button
    lateinit var binv: Button
    lateinit var b0: Button
    lateinit var b9: Button
    lateinit var b8: Button
    lateinit var b7: Button
```

```
lateinit var b6: Button
lateinit var b5: Button
lateinit var b4: Button
lateinit var b3: Button
lateinit var b2: Button
lateinit var b1: Button
lateinit var bpi: Button
lateinit var bmul: Button
lateinit var bminus: Button
lateinit var bplus: Button
lateinit var bequal: Button
lateinit var bdot: Button
lateinit var bdiv: Button

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    // initializing all our variables.
    tvsec = findViewById(R.id.idTVSecondary)
    tvMain = findViewById(R.id.idTVprimary)
    bac = findViewById(R.id.bac)
    bc = findViewById(R.id.bc)
    bbrac1 = findViewById(R.id.bbrac1)
    bbrac2 = findViewById(R.id.bbrac2)
    bsin = findViewById(R.id.bsin)
    bcos = findViewById(R.id.bcos)
    btan = findViewById(R.id.btan)
    blog = findViewById(R.id.blog)
    bln = findViewById(R.id.bln)
    bfact = findViewById(R.id.bfact)
    bsquare = findViewById(R.id.bsquare)
```



```
bsqrt = findViewById(R.id.bsqrt)
binv = findViewById(R.id.binv)
b0 = findViewById(R.id.b0)
b9 = findViewById(R.id.b9)
b8 = findViewById(R.id.b8)
b7 = findViewById(R.id.b7)
b6 = findViewById(R.id.b6)
b5 = findViewById(R.id.b5)
b4 = findViewById(R.id.b4)
b3 = findViewById(R.id.b3)
b2 = findViewById(R.id.b2)
b1 = findViewById(R.id.b1)
bpi = findViewById(R.id.bpi)
bmul = findViewById(R.id.bmul)
bminus = findViewById(R.id.bminus)
bplus = findViewById(R.id.bplus)
bequal = findViewById(R.id.bequal)
bdot = findViewById(R.id.bdot)
bdiv = findViewById(R.id.bdiv)

// adding on click listener to our all buttons.
b1.setOnClickListener {
    // on below line we are appending
    // the expression to our text view.
    tvMain.text = (tvMain.text.toString() + "1")
}
b2.setOnClickListener {
    // on below line we are appending
    // the expression to our text view.
    tvMain.text = (tvMain.text.toString() + "2")
}
b3.setOnClickListener {
```

```
// on below line we are appending
// the expression to our text view.
tvMain.text = (tvMain.text.toString() + "3")
}
b4.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "4")
}
b5.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "5")
}
b6.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "6")
}
b7.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "7")
}
b8.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "8")
}
b9.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "9")
}
b0.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "0")
}
bdot.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + ".")
}
bplus.setOnClickListener {
    tvMain.text = (tvMain.text.toString() + "+")
}
bdiv.setOnClickListener {
```

```
        tvMain.text = (tvMain.text.toString() + "/"")
    }
    bbrac1.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + "(")
    }
    bbrac2.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + ")")
    }
    bpi.setOnClickListener {
        // on clicking on pi button we are adding
        // pi value as 3.142 to our current value.
        tvMain.text = (tvMain.text.toString() + "3.142")
        tvsec.text = (bpi.text.toString())
    }
    bsin.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + "sin")
    }
    bcos.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + "cos")
    }
    btan.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + "tan")
    }
    binv.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + "^" + "(-1)")
    }
    bln.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + "ln")
    }
    blog.setOnClickListener {
        tvMain.text = (tvMain.text.toString() + "log")
    }
}
```

```
bminus.setOnClickListener {  
    // on clicking on minus we are checking if  
    // the user has already a minus operation on screen.  
    // if minus operation is already present  
    // then we will not do anything.  
    val str: String = tvMain.text.toString()  
    if (!str.get(index = str.length - 1).equals("-")) {  
        tvMain.text = (tvMain.text.toString() + "-")  
    }  
}  
  
bmul.setOnClickListener {  
    // if mul sign is not present in our  
    // text view then only we are adding  
    // the multiplication operator to it.  
    val str: String = tvMain.text.toString()  
    if (!str.get(index = str.length - 1).equals("*")) {  
        tvMain.text = (tvMain.text.toString() + "*")  
    }  
}  
  
bsqrt.setOnClickListener {  
    if (tvMain.text.toString().isEmpty()) {  
        // if the entered number is empty we are displaying an error message.  
        Toast.makeText(this, "Please enter a valid number..", Toast.LENGTH_SHORT).show()  
    } else {  
        val str: String = tvMain.text.toString()  
        // on below line we are calculation  
        // square root of the given number.  
        val r = Math.sqrt(str.toDouble())  
        // on below line we are converting our double  
        // to string and then setting it to text view.  
        val result = r.toString()
```

```
        tvMain.setText(result)
    }
}
bequal.setOnClickListener {
    val str: String = tvMain.text.toString()
    // on below line we are calling an evaluate
    // method to calculate the value of expressions.
    val result: Double = evaluate(str)
    // on below line we are getting result
    // and setting it to text view.
    val r = result.toString()
    tvMain.setText(r)
    tvsec.text = str
}
bac.setOnClickListener {
    // on clicking on ac button we are clearing
    // our primary and secondary text view.
    tvMain.setText("")
    tvsec.setText("")
}
bc.setOnClickListener {
    // on clicking on c button we are clearing
    // the last character by checking the length.
    var str: String = tvMain.text.toString()
    if (!str.equals("")) {
        str = str.substring(0, str.length - 1)
        tvMain.text = str
    }
}
bsquare.setOnClickListener {
    if (tvMain.text.toString().isEmpty()) {
        // if the entered number is empty we are displaying an error message.
```

```
        Toast.makeText(this, "Please enter a valid number..", Toast.LENGTH_SHORT).show()
    } else {
        // on below line we are getting the expression and then calculating the square of the
number
        val d: Double = tvMain.getText().toString().toDouble()
        // on below line we are calculating the square.
        val square = d * d
        // after calculating the square we
        // are setting it to text view.
        tvMain.setText(square.toString())
        // on below line we are setting
        // the d to secondary text view.
        tvsec.text = "$d2"
    }
}

bfact.setOnClickListener {
    if (tvMain.text.toString().isEmpty()) {
        // if the entered number is empty we are displaying an error message.
        Toast.makeText(this, "Please enter a valid number..", Toast.LENGTH_SHORT).show()
    } else {
        // on below line we are getting int value
        // and calculating the factorial value of the entered number.
        val value: Int = tvMain.text.toString().toInt()
        val fact: Int = factorial(value)
        tvMain.setText(fact.toString())
        tvsec.text = "$value`!"
    }
}

}

}
```

```
fun factorial(n: Int): Int {  
    // this method is use to find factorial  
    return if (n == 1 || n == 0) 1 else n * factorial(n - 1)  
}  
  
fun evaluate(str: String): Double {  
    return object : Any() {  
        // on below line we are creating variable  
        // for tracking the position and char pos.  
        var pos = -1  
        var ch = 0  
  
        // below method is for moving to next character.  
        fun nextChar() {  
            // on below line we are incrementing our position  
            // and moving it to next position.  
            ch = if (++pos < str.length) str[pos].toInt() else -1  
        }  
  
        // this method is use to check the extra space  
        // present in the expression and removing it.  
        fun eat(charToEat: Int): Boolean {  
            while (ch == '.toInt()) nextChar()  
            // on below line we are checking the char pos  
            // if both is equal then we are returning it to true.  
            if (ch == charToEat) {  
                nextChar()  
                return true  
            }  
            return false  
        }  
    }  
}
```

```
// below method is to parse our
// expression and to get the ans
// in this we are calling a parse
// expression method to calculate the value.
fun parse(): Double {
    nextChar()
    val x = parseExpression()
    if (pos < str.length) throw RuntimeException("Unexpected: " + ch.toChar())
    return x
}

// in this method we will only perform addition and
// subtraction operation on the expression.
fun parseExpression(): Double {
    var x = parseTerm()
    while (true) {
        if (eat('+'.toInt())) x += parseTerm() // addition
        else if (eat('-'.toInt())) x -= parseTerm() // subtraction
        else return x
    }
}

// in below method we will perform
// only multiplication and division operation.
fun parseTerm(): Double {
    var x = parseFactor()
    while (true) {
        if (eat('*'.toInt())) x *= parseFactor() // multiplication
        else if (eat('/'.toInt())) x /= parseFactor() // division
        else return x
    }
}
```



```
// below method is use to parse the factor
fun parseFactor(): Double {
    //on below line we are checking for addition
    // and subtraction and performing unary operations.
    if (eat('+'.toInt())) return parseFactor() // unary plus
    if (eat('-'.toInt())) return -parseFactor() // unary minus
    // creating a double variable for ans.
    var x: Double
    // on below line we are creating
    // a variable for position.
    val startPos = pos
    // on below line we are checking
    // for opening and closing parenthesis.
    if (eat('(').toInt()) { // parentheses
        x = parseExpression()
        eat(')').toInt()
    } else if (ch >= '0'.toInt() && ch <= '9'.toInt() || ch == '.'.toInt()) {
        // numbers
        while (ch >= '0'.toInt() && ch <= '9'.toInt() || ch == '.'.toInt()) nextChar()
        // on below line we are getting sub string from our string using start and pos.
        x = str.substring(startPos, pos).toDouble()
    } else if (ch >= 'a'.toInt() && ch <= 'z'.toInt()) {
        // on below function we are checking for the operator in our expression.
        while (ch >= 'a'.toInt() && ch <= 'z'.toInt()) nextChar()
        val func = str.substring(startPos, pos)
        // calling a method to parse our factor.
        x = parseFactor()
        // on below line we are checking for square root.
        x =
        if (func == "sqrt") Math.sqrt(x)
        // on below line we are checking for sin function
```

```
// and calculating sin function using Math class.
else if (func == "sin") Math.sin(
    Math.toRadians(x)
    // on below line we are calculating the cos value
) else if (func == "cos") Math.cos(
    Math.toRadians(x)
    // on below line we are calculating
    // the tan value of our expression.
) else if (func == "tan")
    Math.tan(Math.toRadians(x))
// on below line we are calculating
// log value of the expression.
else if (func == "log")
    Math.log10(x)
// on below line we are calculating
// ln value of expression.
else if (func == "ln") Math.log(x)
// if we get any error then
// we simply return the exception.
else throw RuntimeException(
    "Unknown function: $func"
)
} else {
    // if the condition not satisfy then we are returning the exception
    throw RuntimeException("Unexpected: " + ch.toChar())
}
// on below line we are calculating the power of the expression.
if (eat('^'.toInt())) x = Math.pow(x, parseFactor()) // exponentiation
return x
}
// at last calling a parse for our expression.
}.parse()
```

```
}  
}
```

Main xml code

```
<?xml version="1.0" encoding="utf-8"?>  
<RelativeLayout  
    xmlns:android="http://schemas.android.com/apk/res/android"  
    xmlns:app="http://schemas.android.com/apk/res-auto"  
    xmlns:tools="http://schemas.android.com/tools"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:background="@color/black_shade_3"  
    tools:context=".MainActivity">  
  
    <TextView  
        android:id="@+id/idTVSecondary"  
        android:layout_width="match_parent"  
        android:layout_height="70dp"  
        android:background="@color/black_shade_3"  
        android:gravity="bottom"  
        android:maxLines="1"  
        android:padding="10dp"  
        android:paddingTop="30dp"  
        android:text=""  
        android:textAlignment="viewEnd"  
        android:textColor="@color/white"  
        android:textSize="15sp"  
        tools:ignore="RtlCompat" />  
  
    <TextView  
        android:id="@+id/idTVprimary"  
        android:layout_width="match_parent"
```

```
android:layout_height="100dp"
android:layout_below="@id/idTVSecondary"
android:background="@color/black_shade_3"
android:gravity="bottom"
android:maxLines="1"
android:padding="10dp"
android:text=""
android:textAlignment="viewEnd"
android:textColor="#fff"
android:textSize="50sp"
tools:ignore="RtlCompat" />
```

```
<androidx.cardview.widget.CardView
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_below="@id/idTVprimary"
    android:background="@color/blac_shade_1"
    app:cardCornerRadius="4dp"
    app:cardElevation="2dp">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_weight="7"
    android:background="@color/blac_shade_1"
    android:orientation="vertical">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:weightSum="7">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:layout_weight="1"
```

```
    android:orientation="horizontal">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:orientation="horizontal"
```

```
    android:weightSum="4">
```

```
<Button
```

```
    android:id="@+id/bac"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:layout_margin="3dp"
```

```
    android:layout_weight="1"
```

```
    android:backgroundTint="@color/black_shade_2"
```

```
    android:padding="6dp"
```

```
    android:text="AC"
```

```
    android:textColor="@color/yellow"
```

```
    android:textSize="15sp"
```

```
    tools:targetApi="lollipop" />
```

```
<Button
```

```
    android:id="@+id/bc"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:layout_margin="3dp"
```

```
    android:layout_weight="1"
```

```
        android:backgroundTint="@color/black_shade_2"
        android:padding="6dp"
        android:text="C"
        android:textColor="@color/yellow"
        android:textSize="15sp" />
```

```
<Button
    android:id="@+id/bbrac1"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="("
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
<Button
    android:id="@+id/bbrac2"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text=")"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_weight="1"  
    android:orientation="horizontal">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="horizontal"  
    android:weightSum="5">
```

```
<Button
```

```
    android:id="@+id/bsin"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"  
    android:layout_weight="1"  
    android:backgroundTint="@color/black_shade_2"  
    android:padding="6dp"  
    android:text="sin"  
    android:textAllCaps="false"  
    android:textColor="#ffa500"  
    android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/bcos"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"
```

```
android:layout_weight="1"
android:backgroundTint="@color/black_shade_2"
android:padding="6dp"
android:text="cos"
android:textAllCaps="false"
android:textColor="#ffa500"
android:textSize="15sp" />
```

<Button

```
android:id="@+id/btan"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_margin="3dp"
android:layout_weight="1"
android:backgroundTint="@color/black_shade_2"
android:padding="6dp"
android:text="tan"
android:textAllCaps="false"
android:textColor="#ffa500"
android:textSize="15sp" />
```

<Button

```
android:id="@+id/blog"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_margin="3dp"
android:layout_weight="1"
android:backgroundTint="@color/black_shade_2"
android:padding="6dp"
android:text="log"
android:textAllCaps="false"
android:textColor="#ffa500"
```



```
        android:textSize="15sp" />
```

```
    <Button
```

```
        android:id="@+id/bln"
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="match_parent"
```

```
        android:layout_margin="3dp"
```

```
        android:layout_weight="1"
```

```
        android:backgroundTint="@color/black_shade_2"
```

```
        android:padding="6dp"
```

```
        android:text="ln"
```

```
        android:textAllCaps="false"
```

```
        android:textColor="#ffa500"
```

```
        android:textSize="15sp" />
```

```
    </LinearLayout>
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:layout_weight="1"
```

```
    android:orientation="horizontal">
```

```
    <LinearLayout
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="match_parent"
```

```
        android:orientation="horizontal"
```

```
        android:weightSum="5">
```

```
<Button
    android:id="@+id/bfact"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="x!"
    android:textAllCaps="false"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
<Button
    android:id="@+id/bsquare"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="x²"
    android:textAllCaps="false"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
<Button
    android:id="@+id/bsqrt"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
```

```
        android:backgroundTint="@color/black_shade_2"
        android:padding="6dp"
        android:text="√"
        android:textColor="#ffa500"
        android:textSize="15sp" />
```

```
<Button
    android:id="@+id/binv"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="1/x"
    android:textAllCaps="false"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
<Button
    android:id="@+id/bdiv"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="÷"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_weight="1"  
    android:orientation="horizontal">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="horizontal"  
    android:weightSum="4">
```

```
<Button
```

```
    android:id="@+id/b7"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"  
    android:layout_weight="1"  
    android:backgroundTint="@color/black_shade_2"  
    android:padding="6dp"  
    android:text="7"  
    android:textColor="#fff"  
    android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/b8"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"
```

```
        android:layout_weight="1"
        android:backgroundTint="@color/black_shade_2"
        android:padding="6dp"
        android:text="8"
        android:textColor="#fff"
        android:textSize="15sp" />
```

```
<Button
    android:id="@+id/b9"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="9"
    android:textColor="#fff"
    android:textSize="15sp" />
```

```
<Button
    android:id="@+id/bmul"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="×"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_weight="1"  
    android:orientation="horizontal">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="horizontal"  
    android:weightSum="4">
```

```
<Button
```

```
    android:id="@+id/b4"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"  
    android:layout_weight="1"  
    android:backgroundTint="@color/black_shade_2"  
    android:padding="6dp"  
    android:text="4"  
    android:textColor="#fff"  
    android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/b5"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"
```

```
        android:layout_weight="1"
        android:backgroundTint="@color/black_shade_2"
        android:padding="6dp"
        android:text="5"
        android:textColor="#fff"
        android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/b6"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="6"
    android:textColor="#fff"
    android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/bminus"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="-"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_weight="1"  
    android:orientation="horizontal">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="horizontal"  
    android:weightSum="4">
```

```
<Button
```

```
    android:id="@+id/b1"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"  
    android:layout_weight="1"  
    android:backgroundTint="@color/black_shade_2"  
    android:padding="6dp"  
    android:text="1"  
    android:textColor="#fff"  
    android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/b2"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"
```



```
        android:layout_weight="1"
        android:backgroundTint="@color/black_shade_2"
        android:padding="6dp"
        android:text="2"
        android:textColor="#fff"
        android:textSize="15sp" />
```

```
<Button
    android:id="@+id/b3"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="3"
    android:textColor="#fff"
    android:textSize="15sp" />
```

```
<Button
    android:id="@+id/bplus"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="+"
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_weight="1"  
    android:orientation="horizontal">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="horizontal"  
    android:weightSum="4">
```

```
<Button
```

```
    android:id="@+id/bpi"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"  
    android:layout_weight="1"  
    android:backgroundTint="@color/black_shade_2"  
    android:padding="6dp"  
    android:text="π"  
    android:textColor="#fff"  
    android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/b0"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:layout_margin="3dp"
```

```
android:layout_weight="1"
android:backgroundTint="@color/black_shade_2"
android:padding="6dp"
android:text="0"
android:textColor="#fff"
android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/bdot"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="."
    android:textColor="#fff"
    android:textSize="15sp" />
```

```
<Button
```

```
    android:id="@+id/bequal"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="3dp"
    android:layout_weight="1"
    android:backgroundTint="@color/black_shade_2"
    android:padding="6dp"
    android:text="="
    android:textColor="#ffa500"
    android:textSize="15sp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

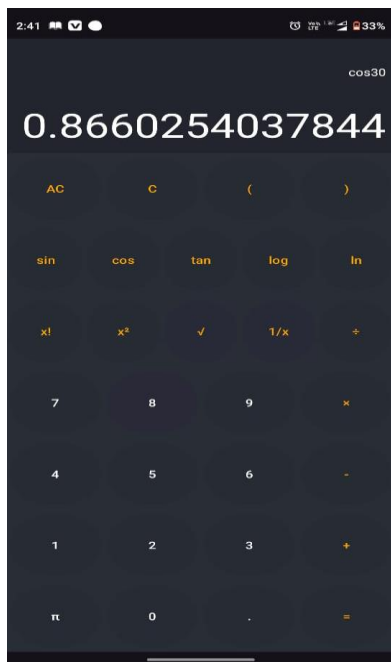
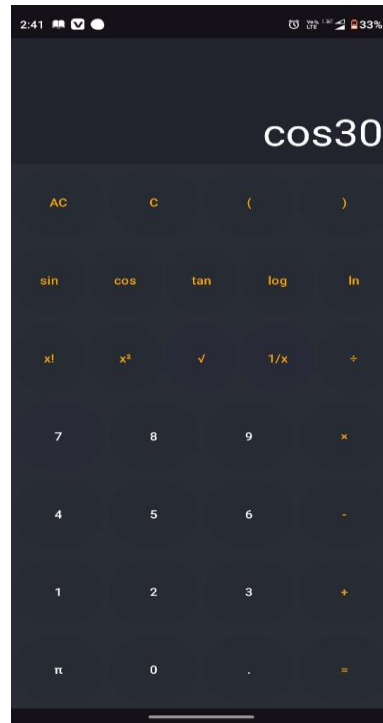
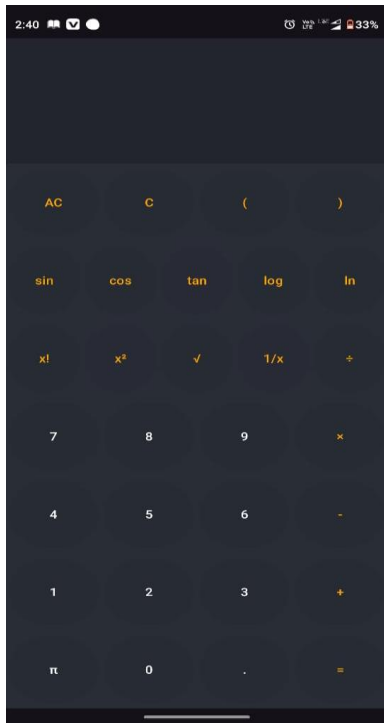
```
</LinearLayout>
```

```
</LinearLayout>
```

```
</androidx.cardview.widget.CardView>
```

```
</RelativeLayout>
```

5.Output



6.Conclusion

In conclusion, the scientific calculator stands as a fundamental tool in the realms of academia, engineering, and scientific research. Its multifaceted functionality empowers users to tackle complex mathematical equations, trigonometric calculations, logarithmic functions, and more with precision and efficiency. Despite advancements in digital technology, the enduring appeal of the scientific calculator lies in its portability, ease of use, and cost-effectiveness. Whether utilized by students mastering foundational concepts or by professionals navigating intricate calculations, the scientific calculator remains an indispensable asset, seamlessly blending traditional mathematical principles with modern convenience.

7.References

[Ionic Web Development - Build Your Mobile/Web Solution](#)

[How to Make a Scientific Calculator Android App using Android Calculator Design – How to Prototype a Functioning Calculator](#)